



THE LONDON SCHOOL
OF ECONOMICS AND
POLITICAL SCIENCE ■

September 2026 Exam

EC400

Static Optimization and Fixed Points

Suitable for all candidates

Instructions to candidates

This paper contains 4 questions. Answer all questions.

Answers should be justified by showing work. Please write your answers in dark ink (black or blue) only.

Only answers written in an answer booklet will be sent for marking.

Time Allowed	Reading Time:	<i>None</i>
	Writing Time:	<i>90 minutes</i>

You are supplied with:	<i>No additional materials</i>
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You may also use:	<i>No additional materials</i>
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Calculators:	<i>Calculators are not allowed in this exam</i>
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If during this exam you feel you need to make additional assumptions in order to proceed, then please do so by stating your assumptions clearly.

1. For $(a, b) \in \mathbb{R}^2$, consider the following quadratic form

$$Q(\mathbf{x}) = 2(a - b)x_1x_2 + a(x_1^2 + x_2^2).$$

- Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is, find the symmetric matrix A satisfying $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$. Then, find all the principal minors of A . **(4 marks)**
 - Define what it means for A to be *indefinite*, *positive semidefinite* and *negative semidefinite*. Then, find all the values for a and b such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*? **(12 marks)**
 - Plot the pairs (a, b) for which $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite* in $\mathbb{R}^2$. **(4 marks)**
2. Consider the function $f(\mathbf{x}) = x_1^2 x_2 e^{-x_2}$ defined on some convex set U in $\mathbb{R}_{++}^2$.
- Define what it means for f to be *quasi-concave* and *quasi-convex*. **(6 marks)**
 - Is f *quasi-concave* or *quasi-convex*? Hint: Use counterexamples to show that a property fails. **(14 marks)**
3. For $a > 0$, consider the problem:

$$\max_{x,y} xy^2 \quad \text{s.t.} \quad x + 2y \leq a, \quad x \geq \frac{1}{a}, \quad y \geq 0.$$

- For what values of $a > 0$ is *Slater condition* satisfied? For such values of a , do *Kuhn-Tucker necessary first order conditions* identify a solution to the constrained optimization problem? **(10 marks)**
 - Write down the *Kuhn-Tucker necessary first order conditions* that must be satisfied by the solution of the constrained optimization problem. **(6 marks)**
 - Solve the constrained optimization problem for all values of a satisfying the restrictions found in part (a). **(18 marks)**
 - Suppose the restrictions on a found in part (a) hold. Use the *Envelope Theorem* to express the slope of value function with respect to the parameter a in terms of the solution $(x(a), y(a), \lambda(a))$ found in part (c). **(6 marks)**
4. Consider the correspondence $\mathbf{h} : [0, 4]^2 \rightrightarrows [0, 4]^2$, where $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}))$ and

$$h_1(\mathbf{x}) = \begin{cases} [0, 2] & \text{for } x_2 \in [0, 1) \\ [0, 4] & \text{for } x_2 = 1 \\ [2, 4] & \text{for } x_2 \in (1, 4] \end{cases}$$

$$h_2(\mathbf{x}) = 4 - x_1 \text{ for } x_1 \in [0, 4].$$

Argue why *Kakutani fixed point theorem* applies. Then, find all the *fixed points* of this correspondence. **(20 marks)**